Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Customized Powder-Based Coating Processes for Industrial Applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen (√), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical Materials		Mechanical Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Surface Treatment	Coating/Method Technologies	Metal Materials	Functional Materials
Concept of Proposed Technology	The proposed technology focuses on customized powder coating processes for the targeted improvement of coatings on porous and compact substrates. By adapting powder-based coating approaches to specific challenges within an industrial process chain, coating composition, surface topography, and microstructure can be precisely controlled, enabling improved homogeneity, reproducibility, and functional performance. Both wet and dry coating strategies are available and can be tailored to the partner's process requirements, with the aim of improving process stability, scalability, and the potential for increased process integration or automation. Where required, post-deposition thermal treatments are applied to ensure reliable metallurgical or functional bonding between coating and substrate, thereby stabilizing coating quality for industrial implementation.			

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Rationale for Joint R&D Project	The rationale for this joint R&D project is based on established expertise in powder-based coating technologies and the experience in transferring such technologies into application-oriented contexts, including electrode materials for alkaline water electrolysis and water treatment. However, challenges related to coating quality, homogeneity, reproducibility, and process robustness are often highly specific to individual industrial process chains and cannot be fully addressed without close interaction with industrial partners. A joint R&D framework enables the systematic adaptation and further development of customized coating processes in direct alignment with partner-specific requirements and operating conditions. Through collaborative development, testing, and iterative optimization, the project provides a platform for mutual knowledge exchange, validation of improved coating concepts, and the exploration of scalable and transferable solutions, thereby supporting long-term international cooperation with Korean partners.
Key Words	1. coating technology 2. surface modification 3. coating homogeneity 4. coating reproducibility 5. powder metallurgy

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	The objective of this R&D project is to further develop and adapt powder-based coating technologies to address partner-specific challenges within existing coating and manufacturing processes. The focus lies on the customization and improvement of coating routes for new or evolving material systems, such as for example functional or catalytic materials provided by project partners, with the aim of enhancing coating homogeneity, reproducibility, and process robustness. By systematically exploring and tailoring wet and dry coating approaches, the project seeks to establish application-oriented coating methodologies that support improved process stability, scalability, and informed decision-making for industrial implementation.			
R&D Target Specifications	The target specifications of this R&D project focus on the reliable and improved application of partner-specific material systems using customized powder-based coating processes. Key measurable parameters include coating thickness typically in the range of approximately 10–200 µm, a homogeneous coating, and controlled microstructural characteristics such as coating uniformity, composition, and effective surface area. Further targets include improved reproducibility of coating quality, assessment of process scalability.			
R&D Plan	The R&D project is planned for a duration of 36 months and follows a flexible, problem-driven approach aimed at identifying, analyzing, and addressing partner-specific challenges in powder-based coating processes. The project is designed to support industrial partners in improving coating quality, homogeneity, reproducibility, and process robustness at critical points within their existing processes. **Phase 1: Identification of partner-specific coating challenges (Months 1–3)** In close collaboration with the project partners, existing coating processes and quality limitations will be analyzed. Target applications, performance expectations, and relevant operating conditions will be defined. Partner-provided materials will be assessed with respect to their current coating behavior, and suitable substrates will be selected to reflect realistic industrial conditions. This phase establishes a shared understanding of the key challenges to be addressed within the project. **Phase 2: Evaluation and development of coating methodologies (Months 4–28)** Based on the identified challenges, suitable wet and dry powder coating approaches will be selected, adapted, and further developed to improve coating performance. This phase includes the optimization of coating parameters, formulation and adjustment of suspensions where required, and the controlled application of coatings to substrates, with a focus on improving coating uniformity and process stability. Where necessary, post-deposition thermal treatments such as debinding, annealing, and sintering will be applied to improve microstructure, phase composition, and coating–substrate bonding. These treatments may be conducted under oxidizing (air), inert (N_2, Ar), or reducing (H_2) atmospheres, under high vacuum, and, where appropriate, using pressure-assisted techniques such as hot pressing or spark plasma sintering. In addition, adaptations of both the substrate and the coating powder may be implemented to address specific process limitations. This includes substrate pre-treatments such as pressing, rolling, bending, stamping, laser cutting, waterjet cutting, or electrical discharge machining (EDM), as well as powder modification through mixing or milling. The objective of this phase is to establish robust, reproducible, and transferable coating solutions tailored to the partner's needs. **Phase 3: Parallel characterization and analysis (Months 1–36)** Comprehensive characterization will be carried out throughout the project to support root-cause analysis and guide process improvements. This includes the analysis of raw partner materials, suspensions (if applicable), and coated samples. Key analytical methods comprise microstructural analysis (SEM), compositional analysis (EDX, ICP, CNOHS gas analysis), X-ray diffraction (XRD), surface area and roughness characterization (BET,			

Hg porosimetry, profilometry), powder characterization (particle size distribution, apparent and tap density, powder flow behavior), and wetting characterization using tensiometry.
Depending on the identified challenges, selected mechanical (tensile, compression, bending tests, RFDA), thermal (DSC, DTA, TG, dilatometry), electrochemical, and physical properties (electrical and thermal conductivity, pressure drop, magnetic properties) can be evaluated to establish process–structure–property relationships and to identify the key parameters limiting coating performance.

Phase 4: Partner-led validation (Months 20–36)
Coating solutions demonstrating improved performance and process stability will be prepared and applied to substrates to enable application-oriented evaluation by the project partners. During this phase, the partners will actively test the optimized coatings within their own experimental setups, processes, or pilot environments under relevant operating conditions. The results obtained will be evaluated, and technical feedback from the partners will be used to assess the achieved improvements, identify remaining limitations, and, where appropriate, define further optimization steps or pathways toward industrial implementation.

Key Milestones

- **M1 (Month 14):** Identification of key coating-related challenges and demonstration of first improved coating solutions
- **M2 (Month 24):** Validation of improved process stability, homogeneity, and reproducibility for selected coating routes
- **M3 (Month 30):** Testing and assessment of optimized coating solutions, providing a basis for further optimization or potential industrial implementation

The proposed R&D plan represents an initial framework and will be flexibly adapted in close cooperation with the project partner to reflect specific requirements, priorities, and application needs.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Fraunhofer Institute for Manufacturing Technology and Advanced Materials (IFAM) has long-standing expertise in powder metallurgy and advanced coating technologies for functional materials. For more than two decades, the institute has been actively engaged in the development and application of coating processes for porous substrates and complex material systems. These activities encompass the tailored deposition of metallic and functional powders, the control of microstructure and porosity, and the integration of coatings into application-oriented components. Within the joint R&D project, Fraunhofer IFAM contributes extensive know-how in process development, materials characterization, and the optimization of coating–substrate interactions, providing a strong foundation for collaborative R&D and industrial implementation.
Expected R&R of Korean SME Partner	The Korean SME is expected to play a key role in the joint R&D project by contributing its expertise in areas such as material development, coating technologies, or related fields. The partner will provide application-specific functional and catalytic materials and define relevant performance requirements, operating conditions, and existing process constraints. In parallel, the SME may further develop or newly design its material systems during the course of the project in order to enhance functional performance. In close collaboration with Fraunhofer IFAM, these materials will be evaluated and adapted for deposition onto suitable substrates, with the aim of improving coating quality and application performance. The Korean SME will be actively involved in application-oriented testing and evaluation of the optimized coatings under relevant conditions, provide technical feedback to support iterative process improvement, and contribute resources in terms of materials, technical know-how, and personnel. Continuous collaboration and knowledge exchange will ensure alignment of the developed coating solutions with practical industrial needs.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The joint R&D project is expected to enable the industrial partner to achieve either improved or newly developed coating solutions that demonstrate enhanced and more reliable performance when applied to relevant substrates, or to gain a clear understanding of the achievable performance and limitations of customized powder-coated systems. Through the collaboration, coating processes will be systematically adapted and optimized for the partner's material systems, generating validated knowledge on coating behavior, process–structure–property relationships, and critical process parameters. These outcomes will support informed decision-making with respect to further material development, process improvement, and potential commercialization, while reducing development risks and strengthening the partner's technological competitiveness.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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